Introduction to Operations Research, Deterministic Models

Lecture 4: Simplex Methods

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Recap: basic and nonbasic variables

For an LP in the general form with n decision variables and m functional constraints.
Basic and nonbasic variables:

- Each variable is designated as either a nonbasic variable, or a basic variable.
- The number of nonbasic variables equals n. The value of nonbasic variables are equal to 0.
- The number of basic variables is m . The value of basic variables are obtained as the simultaneous solution of the system of equations.

Adjacent basic solutions:

- Two CPF solutions are **adjacent** to each other if they share $n-1$ constraint boundaries.
- Two BF solutions are **adjacent** if they share <u>all but one</u> nonbasic variables, and consequently, they share <u>all but one</u> basic variables.

Exercise

Label each of the following statements as true or false.

(a) For minimization problems with at least one optimal solution, if the objective function evaluated at a CPF solution is no larger than its value at every adjacent CPF solution, then that solution is optimal.

(b) Only CPF solutions can be optimal, so the number of optimal solutions cannot exceed the number of CPF solutions.

(c) The best CPF solution is always an optimal solution.

(d) If multiple optimal solutions exist, then an optimal CPF solution may have an adjacent CPF solution that also is optimal.

(e) A BF solution to an LP in the general form with n decision variables may have more than n variables with value 0.

(f) A BF solution to an LP in the general form with n decision variables may have more than n nonbasic variables.

In this lecture, you'll learn

- Simplex methods for non-general LP models
  - artificial-variable technique
  - big-M method
  - two-phase method
- Simplex method in matrix form
- Introduction of Sensitivity analysis

Simplex methods for non-general LP models

Simplex methods for non-general LP models

The standard simplex method is performed for LP model with the following features:

1. The objective function is to be maximized
2. The functional constraints are in $\leq$ form with nonnegative right-hand sides.
3. The variables have nonnegativity constraints.

What if any of these features is not satisfied?

• The minimization $Z = \sum_{j=1}^{n} c_j x_j$ can be transformed into maximizing $Z' = \sum_{j=1}^{n} (-c_j) x_j$.

• The serious problem resulted from the absence of other features lies in identifying an initial BF solution.

• **Artificial-variable technique** is used to enable obtaining an initial BF solution so that the simplex method can proceed.

Idea of Artificial-variable technique

1. **Introduce an artificial variable** into each functional constraint that does not meet the required feature, which enlarges the feasible region unless the artificial variable is equal to 0. This leads to a revised LP problem and it is then easy to identify an initial BF solution for the revised problem.

2. **Drive the values of the artificial variables to 0** by using special methods (big-M method and two-phase method) to enable the simplex method to convert the revised problem back into the original problem, and identify a BF solution for the original problem.

3. The simplex method can proceed with that BF solution to solve the original problem.

Artificial-variable: equality constraints

A modified Wyndor problem with one equality constraint:

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad \leq 4 \\ & \qquad\quad 2x_2 \leq 12 \\ & 3x_1 + 2x_2 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2. \end{array}$$

Introducing a nonnegative artificial variable $\bar{x}_5$ into the third equation leads to

$$3x_1 + 2x_2 + \bar{x}_5 = 18, \qquad \bar{x}_5 \geq 0$$

which **enlarges the feasible region** except when $\bar{x}_5 = 0$.
We could identify an initial BF $(0, 0, 4, 12, 18)$ for the revised problem

$$\boxed{\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad + x_3 \qquad\quad = 4 \\ & \qquad 2x_2 \quad + x_4 \quad = 12 \\ & 3x_1 + 2x_2 \qquad\quad + \bar{x}_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4, 5. \end{array}}$$

Artificial-variable: $\geq$ functional constraints

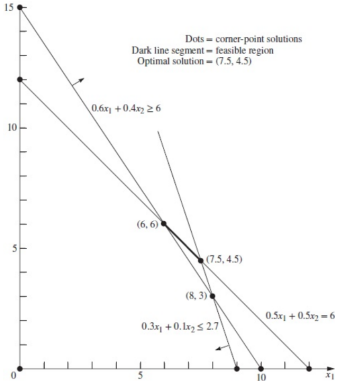
If we multiply the both sides of $\geq$ functional constraints by -1, we obtain an $\leq$ functional constraints but with **negative RHS**.

So a proper method is to first transform $\geq$ functional constraints into an equality by subtracting a surplus variable, and thus add an artificial variable to enable the identification of an initial BF.

Example:

$$\begin{array}{ll} \text{maximize} & Z = 4x_1 + 5x_2 \\ \text{subject to} & 3x_1 + x_2 \leq 27 \\ & 5x_1 + 5x_2 = 60 \\ & 6x_1 + 4x_2 \geq 60 \\ & x_j \geq 0, \text{ for } j = 1, 2. \end{array}$$

$$\begin{aligned} & 6x_1 + 4x_2 \geq 60 \\ \rightarrow\ & 6x_1 + 4x_2 - x_5 = 60, x_5 \geq 0 \\ \rightarrow\ & 6x_1 + 4x_2 - x_5 + \bar{x}_6 = 60, x_5 \geq 0, \bar{x}_6 \geq 0 \end{aligned}$$



Artificial-variable: $\geq$ functional constraints

Original problem:

$$\begin{array}{ll} \text{minimize} & Z = 4x_1 + 5x_2 \\ \text{subject to} & 3x_1 + x_2 \leq 27 \\ & 5x_1 + 5x_2 = 60 \\ & 6x_1 + 4x_2 \geq 60 \\ & x_j \geq 0, \text{ for } j = 1, 2. \end{array}$$

Revised problem:

$$\begin{array}{ll} \text{minimize} & Z = 4x_1 + 5x_2 \\ \text{subject to} & 3x_1 + x_2 + \mathbf{x_3} = 27 \\ & 5x_1 + 5x_2 + \bar{x}_4 = 60 \\ & 6x_1 + 4x_2 - x_5 + \bar{x}_6 = 60 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 5. \\ & \bar{x}_j \geq 0, \text{ for } j = 4, 6. \end{array}$$

- slack variable: $\mathbf{x}_3$
- surplus variable: x_5
- artificial variable: $\bar{x}_4$, $\bar{x}_6$

We could identify an initial BF $(0, 0, 27, 60, 0, 60)$ for the revised problem

Artificial-variable: negative RHS

• Nonnegative right-hand sides for all the functional constraints enables the simplex method to set values of the initial basic variables as RHS to satisfy nonnegativity constraints.

When we have a functional constraint with a negative RHS, e.g., $\boxed{3x_1 - x_2 \leq -2}$

• We first multiply through both sides by -1 to reverse the sign of RHS which also reverses the direction of the inequality; thus leads to a $\geq$ functional constraint, e.g., $\boxed{-3x_1 + x_2 \geq 2}$

• We then introduce the needed surplus variable, and artificial variable to enable the identification of an initial BF.

Variables without nonnegativity constraints

- **Case 1:** variables with a lower bound.

Suppose x_j is allowed to be negative, with a lower bound: $x_j \geq L_j$ with a constant L_j.

Conversion: introduce a new variable $\boxed{x'_j = x_j - L_j}$ with $x'_j \geq 0$, and construct an equivalent LP by replacing x_j in the original LP with $x'_j + L_j$.

- **Case 2:** Variables without a lower bound

Suppose x_j is allowed to be negative, without a lower bound.

Conversion: introduce two new nonnegative variables x_j^+ and x_j^-, and let $\boxed{x_j = x_j^+ - x_j^-}$ with $x_j^+, x_j^- \geq 0$ to construct an equivalent LP.

Interpretation: Each BF solution for the new form of the model necessarily has the property that either $x_j^+ = 0$, or $x_j^- = 0$. Thus x_j^+ represents the positive part of x_j and x_j^- its negative part.

The Big-M method: general procedure

1. Revise the LP by introducing artificial variables, surplus variables, slack variables as needed, which provides an initial BF solution for the revised problem.

2. Construct the artificial problem by further subtracting the objective function by the product of an extremely large constant M and each artificial variable, in order to drive the artificial variables to 0 via the follow-up simplex procedure.

3. Since the artificial variables are basic variables in the initial BF solution to the artificial problem, we need to perform algebraic operations to algebraically eliminate the artificial variables in row 0 so that only nonbasic variables are allowed in row 0.

4. Apply the simplex method to the artificial problem until all of the artificial variables have been driven to nonbasic variables with value zero. This yields a BF solution with all artificial variables being 0.

5. Apply the simplex method to the original problem with the last obtained BF solution until it reaches an optimal solution.

The Big-M method: example

Original problem

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 \\ \text{subject to} & x_1 \qquad\quad \leq 4 \\ & \qquad\quad 2x_2 \leq 12 \\ & 3x_1 + 2x_2 = 18 \\ & x_1 \geq 0, x_2 \geq 0 \end{array}$$

Artificial problem

$$\begin{array}{ll} \text{maximize} & Z = 3x_1 + 5x_2 - M\bar{x}_5 \\ \text{subject to} & x_1 \qquad\quad + x_3 = 4 \\ & \qquad 2x_2 \quad + x_4 = 12 \\ & 3x_1 + 2x_2 \qquad\quad + \bar{x}_5 = 18 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 4. \\ & \bar{x}_5 \geq 0. \end{array}$$

1. initial BF $(0, 0, 4, 12, 18)$
2. new objective equation that only involves nonbasic variables:

$$Z = (3x_1 + 5x_2) - M(18 - 3x_1 - 2x_2) = -18M + (3M + 3)x_1 + (2M + 5)x_2$$

3. Perform simplex method to the artificial problem using simplex tableaux

The Big-M method: example

Iteration	Basic variable	row	Coefficient						RHS	ratio
			Z	x_1	x_2	x_3	x_4	$\bar{x}_5$		
0	Z	(0)	1	-(3M+3)	-(2M+5)	0	0	0	-18M	
	x_3	(1)	0	1	0	1	0	0	4	
	x_4	(2)	0	0	2	0	1	0	12	
	$\bar{x}_5$	(3)	0	3	2	0	0	1	18	

Two-phase method

The big-M method becomes quite computationally complicated for more than one artificial variables.

Original problem

$$\begin{array}{ll} \text{minimize} & 4x_1 + 5x_2 \\ \text{subject to} & 3x_1 + x_2 \leq 27 \\ & 5x_1 + 5x_2 = 60 \\ & 6x_1 + 4x_2 \geq 60 \\ & \bar{x}_1 \geq 0, x_2 \geq 0. \end{array}$$

Big-M problem

$$\begin{array}{ll} \text{maximize} & -4x_1 - 5x_2 - M(\bar{x}_4 + \bar{x}_6) \\ \text{subject to} & 3x_1 + x_2 + \mathbf{x_3} = 27 \\ & 5x_1 + 5x_2 + \bar{x}_4 = 60 \\ & 6x_1 + 4x_2 - x_5 + \bar{x}_6 = 60 \\ & x_j \geq 0, \text{ for } j = 1, 2, 3, 5. \\ & \bar{x}_j \geq 0, \text{ for } j = 4, 6. \end{array}$$

By dividing by M on the objective, we obtain $\dfrac{Z}{M} = \underbrace{-\dfrac{4}{M}x_1 - \dfrac{5}{M}x_2}_{\text{negligible}} - (\bar{x}_4 + \bar{x}_6).$

Two-phase method

Phase-1 problem

$$
\begin{array}{ll}
\text{maximize} & -\bar{x}_4 - \bar{x}_6 \\
\text{subject to} & 3x_1 + x_2 + \mathbf{x_3} = 27 \\
& 5x_1 + 5x_2 + \bar{x}_4 = 60 \\
& 6x_1 + 4x_2 - x_5 + \bar{x}_6 = 60 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 5. \\
& \bar{x}_j \geq 0, \text{ for } j = 4, 6.
\end{array}
$$

Augmented real problem

$$
\begin{array}{ll}
\text{maximize} & -4x_1 - 5x_2 \\
\text{subject to} & 3x_1 + x_2 + \mathbf{x_3} = 27 \\
& 5x_1 + 5x_2 = 60 \\
& 6x_1 + 4x_2 - x_5 = 60 \\
& x_j \geq 0, \text{ for } j = 1, 2, 3, 5.
\end{array}
$$

In Phase 1: solve the artificial problem until $\bar{x}_4 = 0$ and $\bar{x}_6 = 0$, which yield the initial BF solution for the augmented real problem;

In Phase 2: Use simplex method to solve the real problem with the initial BF solution obtained in phase 1.

Two-phase method: Phase 1

Iteration	Basic variable	row	Coefficient							RHS	ratio
			Z	x_1	x_2	x_3	$\bar{x}_4$	x_5	$\bar{x}_6$		
0	Z	(0)	1	-11	-9	0	0	1	0	-120	
	x_3	(1)	0	3	1	1	0	0	0	27	
	$\bar{x}_4$	(2)	0	5	5	0	1	0	0	60	
	$\bar{x}_6$	(3)	0	6	4	0	0	-1	1	60	
...	...	...	...								
3	Z	(0)	1	0	0	0	1	0	1	0	
	x_1	(1)	0	1	0	0	-4	-5	5	6	
	x_3	(2)	0	0	0	1	$\frac{3}{5}$	1	-1	3	
	x_2	(3)	0	0	1	0	6	5	-5	6	

$(6, 6, 3, 0, 0, 0)$ yields a feasible BF solution $(6, 6, 3, 0)$ for the real problem.

Two-phase method: Phase 2

Iteration	Basic variable	row	Coefficient							RHS
			Z	x_1	x_2	x_3	$\bar{x}_4$	x_5	$\bar{x}_6$	
Final phase 1 tableau	Z	(0)	1	0	0	0	1	0	1	0
	x_1	(1)	0	1	0	0	-4	-5	5	6
	x_3	(2)	0	0	0	1	$\frac{3}{5}$	1	-1	3
	x_2	(3)	0	0	1	0	6	5	-5	6
drop $\bar{x}_4$ and $\bar{x}_5$, substitute phase 2 objective	Z	(0)	1	4	5	0		0		0
	x_1	(1)	0	1	0	0		-5		6
	x_3	(2)	0	0	0	1		1		3
	x_2	(3)	0	0	1	0		5		6
Phase 2 tableau	Z	(0)	1	0	0	0		-5		-54
	x_1	(1)	0	1	0	0		-5		6
	x_3	(2)	0	0	0	1		1		3
	x_2	(3)	0	0	1	0		5		6

Starting at the BF solution $(6, 6, 3, 0)$ in Phase 2 tableau, we apply the simplex method to

Simplex method in matrix form

LP in matrix form

General form

$$\boxed{\begin{array}{ll} \text{maximize} & Z = \mathbf{c}^\top \mathbf{x} \\ \text{subject to} & \mathbf{A}\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \end{array}}$$

Equation system in Simplex tableau: $\begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_s \end{pmatrix} = \begin{pmatrix} 0 \\ \mathbf{b} \end{pmatrix}$

<u>At any</u> of the simplex algorithm, **a new BF solution** is obtained by appropriate row operations (left matrix multiplication) such that

$$\begin{pmatrix} ? \end{pmatrix} \begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_s \end{pmatrix} = \begin{pmatrix} 1 & 0 & ? \\ 0 & \mathbf{I} & ? \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x}_B \\ \mathbf{x}_N \end{pmatrix}$$

where $\mathbf{x}_B \in \mathbb{R}^m$ represent the **vector of basic variables**, and $\mathbf{x}_N \in \mathbb{R}^n$ represent the **vector of non-basic variables**

LP in matrix form

For the new BF solution, let

- $\mathbf{x}_B \in \mathbb{R}^m$ represent the **vector of basic variables**, and $\mathbf{x}_N \in \mathbb{R}^n$ represent the **vector of non-basic variables**
- $\mathbf{B} \in \mathbb{R}^{m \times m}$ represent **the basis matrix** with columns corresponding to coefficients of the basic variables, and $\mathbf{N}$ represent the matrix with columns corresponding to coefficients of the nonbasic variables.

Observe that for the BF solution introduced by the simplex method, the basis matrix $\mathbf{B}$ is always nonsingular, and $\begin{pmatrix} \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} \mathbf{B} & \mathbf{N} \end{pmatrix} \begin{pmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{pmatrix} = \mathbf{b}$.

Thus,

$$\mathbf{B}^{-1} \begin{pmatrix} \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{B}^{-1}\mathbf{N} \end{pmatrix} \begin{pmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{pmatrix} = \mathbf{B}^{-1}\mathbf{b}$$

Row operations of the coefficients of constraints within the simplex method can be interpreted as $\mathbf{B}^{-1}$ multiplying $\begin{pmatrix} \mathbf{A} & \mathbf{I} \end{pmatrix}$.

LP in matrix form

Row operations to the initial equation system is equivalent to following,

$$\begin{pmatrix} 1 & \mathbf{c}_B^\top \mathbf{B}^{-1} \\ 0 & \mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} 1 & \mathbf{c}_B^\top \mathbf{B}^{-1} \\ 0 & \mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} 0 \\ \mathbf{b} \end{pmatrix}$$

Particularly,

$$\mathbf{B}^{-1} \begin{pmatrix} \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} \mathbf{I} & \mathbf{B}^{-1}\mathbf{N} \end{pmatrix} \begin{pmatrix} \mathbf{x}_B \\ \mathbf{x}_N \end{pmatrix}$$

$$\begin{pmatrix} 1 & \mathbf{c}_B^\top \mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = Z + (\mathbf{c}_B^\top \mathbf{B}^{-1}\mathbf{N} - \mathbf{c}_N^\top)\mathbf{x}_N + 0 \cdot \mathbf{x}_B$$

LP in matrix form

$$\begin{pmatrix} 1 & \mathbf{c}_B^\top \mathbf{B}^{-1} \\ 0 & \mathbf{B}^{-1} \end{pmatrix} \begin{pmatrix} 1 & -\mathbf{c}^\top & 0 \\ 0 & \mathbf{A} & \mathbf{I} \end{pmatrix} \begin{pmatrix} Z \\ \mathbf{x} \\ \mathbf{x}_S \end{pmatrix} = \begin{pmatrix} \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{b} \\ \mathbf{B}^{-1} \mathbf{b} \end{pmatrix}$$

	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
0	Z	(0)	1	- $\mathbf{c}^\top$	$\mathbf{0}$	0
	$\mathbf{x}_S$		$\mathbf{0}$	$\mathbf{A}$	$\mathbf{I}$	$\mathbf{b}$
$\cdots$	$\cdots$	$\cdots$	$\cdots$			
	Z	(0)	1	$\mathbf{y}^* \mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^* \mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1} \mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1} \mathbf{b}$

$\mathbf{y}^* \triangleq \mathbf{c}_B^\top \mathbf{B}^{-1}$ denote the coefficients of the slack variable in row (0).

Sensitivity analysis

Sensitivity analysis

After we solve the LP by the simplex method given the constant values of parameters $\mathbf{c}, \mathbf{A}, \mathbf{b}$, what would happen if the parameters take on other possible values?

- does the original optimal solution remain feasible/optimal?
- determine the range of values of the coefficient over which the optimality remains.
- if not optimal, how to find the optimal solution of the new problem utilizing the knowledge of the original LP?

Sensitivity analysis: revised Wyndor example

Original problem

maximize $Z = 3x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

Revised problem

maximize $Z = 4x_1 + 5x_2$

subject to $\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$

$\mathbf{x} \geq \mathbf{0}$

In the original problem, let $\mathbf{c} \triangleq \begin{pmatrix} 3 \\ 5 \end{pmatrix}$, $\mathbf{A} \triangleq \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{pmatrix}$, $\mathbf{b} \triangleq \begin{pmatrix} 4 \\ 12 \\ 18 \end{pmatrix}$.

In the revised problem, let $\bar{\mathbf{c}} \triangleq \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\bar{\mathbf{A}} \triangleq \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 2 & 2 \end{pmatrix}$, $\bar{\mathbf{b}} \triangleq \begin{pmatrix} 4 \\ 24 \\ 18 \end{pmatrix}$.

Recall: original Wyndor example

Iteration	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-3	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	12
	x_5	(3)	0	3	2	0	0	1	18
...	...	...	...						
2	Z	(0)	1	0	0	0	3/2	1	36
	x_3	(1)	0	0	0	1	1/3	-1/3	2
	x_2	(2)	0	0	1	0	1/2	0	6
	x_1	(3)	0	1	0	0	-1/3	1/3	2

In the final tableau, $\mathbf{B} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 2 & 3 \end{pmatrix}$, $\mathbf{B}^{-1} = \begin{pmatrix} 1 & 1/3 & -1/3 \\ 0 & 1/2 & 0 \\ 0 & -1/3 & 1/3 \end{pmatrix}$, $\mathbf{y}^* = \begin{pmatrix} 0 \\ 3/2 \\ 1 \end{pmatrix}$.

Sensitivity analysis: revised Wyndor example

Iteration	Basic variable	row	Coefficient			RHS
			Z	$\mathbf{x}$	$\mathbf{x}_S$	
	Z	(0)	1	$\mathbf{y}^*\mathbf{A} - \mathbf{c}^\top$	$\mathbf{y}^*$	$\mathbf{y}^*\mathbf{b}$
	$\mathbf{x}_B$		0	$\mathbf{B}^{-1}\mathbf{A}$	$\mathbf{B}^{-1}$	$\mathbf{B}^{-1}\mathbf{b}$

With the same row operations, we obtain the final tableau for the revised problem.

Iteration	Basic variable	row	Coefficient						RHS
			Z	x_1	x_2	x_3	x_4	x_5	
0	Z	(0)	1	-4	-5	0	0	0	0
	x_3	(1)	0	1	0	1	0	0	4
	x_4	(2)	0	0	2	0	1	0	24
	x_5	(3)	0	2	2	0	0	1	18
...	.....	...							
2	Z	(0)	1	-2	0	0	3/2	1	36
	x_3	(1)	0	1/3	0	1	1/3	-1/3	2
	x_2	(2)	0	0	1	0	1/2	0	6
	x_1	(3)	0	2/3	0	0	-1/3	1/3	2

Sensitivity analysis: general procedure

After we solve the LP by the simplex method given the constant values of parameters $(\mathbf{c}, \mathbf{A}, \mathbf{b})$, what would be the optimal solutions if some of the parameters take on other possible values?

- **Revision of model:** Make the desired change or changes in the model to be investigated.

- **Revision of final tableau:** Use the simplex in matrix form to determine the resulting changes in the final simplex tableau after the same row operations.

- **Conversion to proper form** of simplex tableau from Gaussian elimination

- **Feasibility test:** checking whether all its basic variable values in the right-side column of the tableau still are nonnegative.

- **Optimality test:** checking whether all its nonbasic variable coefficients in row 0 of the tableau still are nonnegative.

- **Reoptimization:** If this solution fails either test, the new optimal solution can be obtained by using the current tableau from step 3 as the initial simplex tableau with any necessary conversions.

Sensitivity analysis

After we solve the LP by the simplex method given the constant values of parameters $(\mathbf{c}, \mathbf{A}, \mathbf{b})$, what would happen if some of the parameters take on other possible values?

- change in RHS
- change in coefficient of NBV
- introduce of a new variable
- changes in coefficient of BV
- add a new constraint

Reading materials

Section 4.1, 4.2, 4.3, 4.4, *Introduction to Operations Research* by Hillier and Lieberman